

Operating Systems

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SEC-5

WEEK-5

5. Implement a producer-consumer communication system using anonymous pipes where the parent process generates data and the child process consumes it. Measure the communication efficiency.

Develop a program that executes the equivalent of the shell command: `ls -l | grep ".c"` using `fork()`, `pipe()`, `dup2()`, and `exec()` system calls.

Code

```
kolasani_navadeep@Navadee x + v
GNU nano 8.7.1 grep.c *
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
int main()
{
    int p[2];

    pipe(p);

    if (fork() == 0)
    {
        char s[20];

        close(p[1]);
        read(p[0], s, 20);
        printf("%s\n", s);
    }
    else
    {
        close(p[0]);
        write(p[1], "Hello", 6);
        wait(NULL);
    }

    pipe(p);

    if (fork() == 0)
    {
        dup2(p[1], 1);
        execlp("ls", "ls", "-l", NULL);
    }

    if (fork() == 0)
    {
        dup2(p[0], 0);
        execlp("grep", "grep", ".c", NULL);
    }

    wait(NULL);
    wait(NULL);

    return 0;
}
```

output

```
kolasani_navadeep@Navadeep-Vitus:~$ nano grep.c
```

```
kolasani_navadeep@Navadeep-Vitus:~$ gcc grep.c
```

```
kolasani_navadeep@Navadeep-Vitus:~$ ./a.out
```

```
Hello
```

-rw-r--r--	1	kolasani_navadeep	kolasani_navadeep	110	Aug	4	04:12	Current.c
-rw-r--r--	1	kolasani_navadeep	kolasani_navadeep	391	Sep	1	03:06	FCFS.c
-rw-r--r--	1	kolasani_navadeep	kolasani_navadeep	391	Sep	1	03:00	FCFS.coxx
-rw-r--r--	1	kolasani_navadeep	kolasani_navadeep	213	Aug	4	03:33	Null.c
-rw-r--r--	1	kolasani_navadeep	kolasani_navadeep	129	Aug	4	04:17	Processid.c
-rw-r--r--	1	kolasani_navadeep	kolasani_navadeep	617	Sep	1	03:35	RoundRobin.c
-rw-r--r--	1	kolasani_navadeep	kolasani_navadeep	501	Sep	1	03:23	SJF.c
-rw-r--r--	1	kolasani_navadeep	kolasani_navadeep	638	Sep	15	04:51	child.c
-rw-r--r--	1	kolasani_navadeep	kolasani_navadeep	723	Sep	15	04:17	copy.c
-rw-r--r--	1	kolasani_navadeep	kolasani_navadeep	257	Sep	15	04:44	fork.c
-rw-r--r--	1	kolasani_navadeep	kolasani_navadeep	614	Sep	15	04:48	fork1.c
-rw-r--r--	1	kolasani_navadeep	kolasani_navadeep	590	Sep	15	04:57	grep.c
-rw-r--r--	1	kolasani_navadeep	kolasani_navadeep	589	Sep	15	04:16	linux.c
-rw-r--r--	1	kolasani_navadeep	kolasani_navadeep	347	Aug	7	04:11	p1.c
-rw-r--r--	1	kolasani_navadeep	kolasani_navadeep	237	Aug	4	04:03	process.c
-rw-r--r--	1	kolasani_navadeep	kolasani_navadeep	667	Aug	11	03:54	week1.c
-rw-r--r--	1	kolasani_navadeep	kolasani_navadeep	669	Aug	11	04:52	week1.c.save